QN0.3

Problem Description:

You are given an array A of Q integers and Q queries. In each query, you are given an integer i $(1 \le i \le N)$.

Your task is to find the minimum index greater than i $(1 \le i \le N)$ such that:

1. Sum of digits of A_i is greater than the sum of digits of A_j
2. $A_i < A_j$

If there is no answer, then print **-1**.

Constraints

1 < = N, Q ≤ 10^5

1 ≤ Ai ≤ 10^9

1 ≤ Qi ≤N

Input format

- The first line contains two numbers N and Q.
- The next line contains N numbers.
- Next Q lines contain Q queries.

Output format

Print the answer as described in the problem

[redacted]

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 5
62 70 28 62 92
1
5
3
4
2
```

EXPECTED OUTPUT

```
2 -1 4 -1 -1
```

Test Case 2

INPUT (STDIN)

```
6 6
12 71 21 16 91
11
15
31
14
21
```

EXPECTED OUTPUT

```
-1 -1 -1 -1 -1 -1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
if(arr[x]<arr[y])
```

Test Case 2

KEYWORD

```
if(arr2[x]>arr2[y])
```

Code:

```
#include <stdio.h>

int digitSum(int n)
{
int sum = 0;

while(n > 0)
```

```
{
sum = sum + n % 10;
n = n / 10;
}

return sum;
}

int main()
{
int N, Q;
int ar[100005];
int ar2[100005];

printf("Input value:\n");

scanf("%d %d", &N, &Q);

for(int i = 0; i < N; i++)
{
scanf("%d", &ar[i]);
ar2[i] = digitSum(ar[i]);
}

printf("\nOutput:\n");

for(int q = 0; q < Q; q++)
{
int x;
int answer = -1;

scanf("%d", &x);
x = x - 1;

for(int y = x + 1; y < N; y++)
{
if(ar[x] < ar[y])
{
if(ar2[x] > ar2[y])
{
answer = y + 1;
break;
}
}
}

printf("%d", answer);

if(q < Q - 1)
printf(" ");
```

```c
}

printf("\n");

return 0;
}
```

Output;

Test _case 1:

```
5 5
62 70 28 62 92
1
2
5
-1
3
4
2
-1
```

Test _case 2:

```
6 6
12 71 21 16 91
11 15 31 14 21
-1 -1 -1 -1
```

Q.N0 :4

Problem Description:

Dr. Malar was booking a tour package of IRCTC from Chennai to Delhi for his family.

Two of the relatives was interested in joining to this tour.

these two persons are studying engineering in computing technology. only one tickets are remaining in the IRCTC portal.

So, Dr. Malar decided to book one ticket for out of those persons also along with his family members.

she wants to identify the one person out of these persons. he decided to conduct a technical task to identify the right person to travel.

the task was that, implement two stack operations in an array

Can you help them to complete the task?

Constraints

0<n<5 only five elements has to be practiced for this operation

first element pushed into stack1

second element pushed into stack2, likewise elements pushed into alternative stacks vice versa.

Function Description

- Create a data structure *twoStacks* that represents two stacks.
- Implementation of *twoStacks* should use only one array, i.e., both stacks should use the same array for storing elements.
- Following functions must be supported by *twoStacks*.
  push1(int x) –> pushes x to first stack
  push2(int x) –> pushes x to second stack
  pop1() –> pops an element from first stack and return the popped element
  pop2() –> pops an element from second stack and return the popped element
  Implementation of *twoStack* should be space efficient.

Input Format

Single line represents only braces (both curly and square)

Output Format

If the given input balanced then print as Balanced (or) Not Balanced

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
1
2
3
4
5
```

EXPECTED OUTPUT

```
Popped element from stack1 is:5
Popped element from stack2 is:4
```

Test Case 2

INPUT (STDIN)

```
5
4
5
3
4
```

EXPECTED OUTPUT

```
Popped element from stack1 is:4
Popped element from stack2 is:3
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void push1(int x)
```

Test Case 2

KEYWORD

```
void push2(int x)
```

Test Case 3

KEYWORD

```
int pop1()
```

Test Case 4

Code:

```
#include <stdio.h>
#define MAX 5
int arr[MAX];


int top1 = -1;
int top2 = -1;
```